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Language Processing tasks theme

... and NLP applications theme

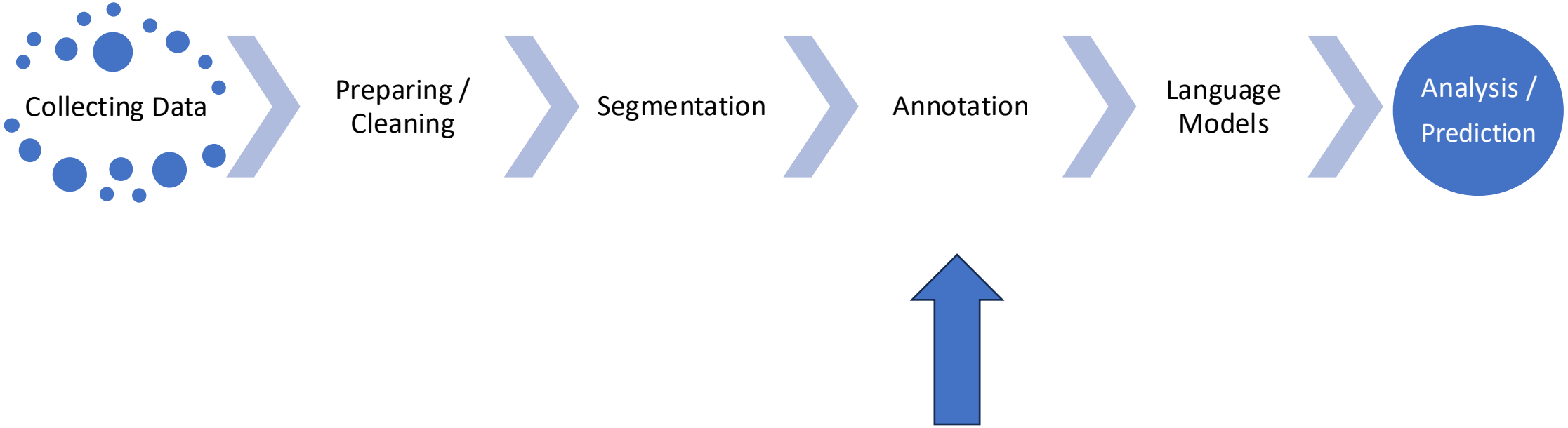
Week 18: Word Sense Disambiguation

SCC453 Natural Language Processing and Language Models

Paul Rayson

With thanks to Andrew Moore for his slides on PyMUSAS

NLP Overview / Pipeline



Types of ambiguity

Remember our introductions to ambiguity in previous weeks?

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- Pragmatic
  - Meaning in context
- Semantic
  - Meaning of the sentence
- Anaphoric
  - Referential
  - What/who does each pronoun refer to?
- Syntactic
  - Structure/grammar
  - How do the words fit together in the sentence?
- Lexical ambiguity
  - Word level
  - Word can have more than one meaning

Lexical ambiguity

- Question:
  - How many senses does *spring* have?
- Answer:
  - A: 3
  - B: 4
  - C: 5
  - D: 31

to arrange the escape of (someone) from prison. **17** (*intr*) *Archaic or poetic.* (of daylight or dawn) to begin to appear. ◆ *n* **18** the act or an instance of springing. **19** a leap, jump, or bound. **20a** the quality of resilience; elasticity. **20b** (*as modifier*): *spring steel.* **21** the act or an instance of moving rapidly back from a position of tension. **22a** a natural outflow of ground water, as forming the source of a stream. **22b** (*as modifier*): *spring water.* **23a** a device, such as a coil or strip of steel, that stores potential energy when it is compressed, stretched, or bent and releases it when the restraining force is removed. **23b** (*as modifier*): *a spring mattress.* **24** a structural defect such as a warp or bend. **25a** (*sometimes cap.*) the season of the year between winter and summer, astronomically from the March equinox to the June solstice in the N hemisphere and from the September equinox to the December solstice in the S hemisphere. **25b** (*as*

etc., running down the inside of the handle of a cricket bat, hockey stick, etc. **29** Also called: **spring line.** *Nautical.* a mooring line, usually one of a pair that cross amidships. **30** a flock of teal. **31** *Architect.* another name for **springing.** [Old English *springan;* related to Old Norse *springa,* Old High German *springan,* Sanskrit *sprhayati* he desires, Old Slavonic *pragu* grasshopper] ▸ **'springless** *adj* ▸ **'spring,like** *adj*

Definition of Collins Englis (21st Century

spring balance *or esp. U.S.* **spring scale** *n* a device in which an object to be weighed is attached to the end of a helical spring, the extension of which indicates the weight of the object on a calibrated scale.

Spring (https://dictionary.cambridge.org)

- *spring* was found in the Cambridge Advanced Learner's Dictionary at the entries listed below.
  - spring (MOVE QUICKLY)
  - spring (APPEAR SUDDENLY)
  - spring (SEASON)
  - spring (CURVED METAL)
  - spring (WATER)
  - box spring
  - spring chicken
  - spring-clean
  - spring greens
  - spring onion
  - spring roll
  - spring from sth
  - spring sth on sb
  - be full of the joys of spring
  - spring to life
  - spring to mind
  - a spring in your step

Word Sense Disambiguation (WSD): overview

- Defined as “the process of automatically deciding the senses of words in context”
- An important area of research for 40+ years (and still not solved, or even started for many of the world’s languages)
- Many applications further along the NLP pipeline and to other real world tasks e.g.
  - machine translation: French translation of ‘free’ in the context of ‘free prisoner’ is *libre* and ‘free gift’ is *gratuit*
  - Information retrieval: finding the correct *jaguar, pandas*, or *apple*, etc

Word Sense Disambiguation (WSD): definitions

- **Homonymy**: distinct meanings with no obvious connection, e.g.
  - *Bank*: ‘edge of a river’ and ‘financial institution’
- **Polysemy**: meanings are related in some way, e.g.
  - *Leak*: ‘let liquid escape’ and ‘give away information’
  - *Mouse*: ‘small furry animal with a tail’ and ‘computer equipment’

Word Sense Disambiguation (WSD): approaches and issues

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- **Knowledge-based**: relies on large manually created knowledge sources e.g. machine readable dictionaries and WordNet, dictionary definition overlap, LDOCE
  - Only as good as the knowledge base
  - Most frequent sense is a very strong baseline
  - Can lexicographers agree on the number / grouping of senses?
- **Supervised machine learning**: use of Wikipedia links or other linked open data source
  - A variety of approaches: decision lists, naïve bayes, maximum entropy, k-nearest neighbour using contextual word embeddings, support vector machines
  - Time consuming to create training data
- **Unsupervised machine learning**: clustering by context similarity
- **Deep learning (neural networks):** relies on extreme scale training corpora to ensure seeing enough training examples even for low frequency senses
- **Domain specific approaches**: one sense per discourse, e.g.
  - *Bat*: 'sports equipment' or 'nocturnal mammal'
  - *Party*: 'politics' or 'entertainment'

Word Sense Disambiguation (WSD): evaluation

- SensEval and SemEval evaluation exercises
  - https://web.eecs.umich.edu/~mihalcea/senseval/
  - https://siglex.org/
  - https://semeval.github.io/
- Lexical sample task, training data provided, untagged data to be tagged by participants systems

Lexical relations and WordNet

- **Synonymy**: senses of two or more words are identical or nearly identical, e.g.
  - *couch/sofa*, *car/automobile*
  - *big/large* (not always replaceable?)
- **Antonymy**: words with opposite meanings, e.g.
  - *long/short*, *fast/slow*, *cold/hot*, *up/down*
- **Hyponym**: subclass relationship, more specific sense, e.g.
  - *car* is a hyponym of *vehicle*
  - *dog* is a hyponym is *animal*
- **Hypernym**: less specific sense, superordinate class

Lexical relations and WordNet

- WordNet lexical database, originally released in 1998, most commonly used resource for sense relations in English and many other languages: https://wordnet.princeton.edu/
- Approx. 120K nouns, 12K verbs, 22K adjectives, 4K adverbs, arranged in synsets (near synonym sets)
- Also see MultiWordNet, Open Multilingual WordNet, BabelNet

Noun

- S: (n) **bass** (the lowest part of the musical range)
- S: (n) **bass**, bass part (the lowest part in polyphonic music)
- S: (n) **bass**, basso (an adult male singer with the lowest voice)
- S: (n) sea bass, **bass** (the lean flesh of a saltwater fish of the family Serranidae)
- S: (n) freshwater bass, **bass** (any of various North American freshwater fish with lean flesh (especially of the genus Micropterus))
- S: (n) **bass**, bass voice, basso (the lowest adult male singing voice)
- S: (n) **bass** (the member with the lowest range of a family of musical instruments)
- S: (n) **bass** (nontechnical name for any of numerous edible marine and freshwater spiny-finned fishes)

Adjective

- S: (adj) **bass**, deep (having or denoting a low vocal or instrumental range) *"a deep voice"; "a bass voice is lower than a baritone voice"; "a bass clarinet"*

Further reading

- Eneko Agirre and Mark Stevenson (2023) Word sense disambiguation. In Ruslan Mitkov (ed.) The Oxford Handbook of Computational Linguistics, second edition.
- Speech and Language Processing (3rd edition 2025) appendix chapter G: https://web.stanford.edu/~jurafsky/slp3/G.pdf
- See Roberto Navigli's guest lecture recording on moodle from last year's SCC413 module (additional materials, also on UCREL YouTube channel)

Any questions so far?

Motivation for semantic analysis

- Understanding the meaning of words and texts
- Various types in the Natural Language Processing literature
  - Named entity recognition
  - Semantic role labelling
  - Word sense disambiguation
  - Summarisation
  - Sentiment analysis
  - Content analysis
  - Semantic field tagging

UCREL Semantic Analysis System (USAS)

What is Semantic Tagging?

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- Semantic field annotation has applications for conceptual or topic tagging:
- *Last_T1.1.1 year_T1.1.1* was_A3+ the_Z5 UK_Z2 's_Z5 second_N4 warmest_O4.6+++ *on_A11.2+ record_A11.2+* ,_PUNC *according_Z5 to_Z5* provisional_T1.3- data_X2.2 from_Z5 the_Z5 Met_S3.1 Office_I2.1/H1c ._PUNC This_Z8 *puts_X2.2-* it_Z8 just_A14 *behind_X2.2-* 2022_N1 ,_PUNC which_Z8 recorded_Q1.2 an_Z5 average_A6.2+ temperature_O4.6 of_Z5 only_A14 0.06C_Z99 higher_N3.7++ ._PUNC

- A3+ = being; A6.2 = comparing; A11.2 = importance; A14 = exclusivisers; H1 = architecture, buildings; I2.1 = business; N1 = numbers; N3.7 = measurement; N4 = linear order; O4.6 = temperature; Q1.2 = documents, writing; S3.1 = relationship; T1.1.1 = Time past; T1.3 = time period; X2.2 = knowledge; Z2 = geographical names; Z5 = grammatical bin; Z8 = pronouns etc; Z99 = unmatched

Multiword expressions: plain sailing?

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- Phrasal verbs
  - *Stubbed out*
- Noun phrases
  - *Riding boots*
  - *Pony nuts*
- Proper names
  - *United States of America*
- *Named entities*
  - *23rd November 1963*
  - *British Broadcasting Corporation*
- Multiword prepositions
  - *In terms of*
  - *As soon as*
- Idiomatic expressions
  - *Spill the beans*
  - *A pain in the neck*

UCREL Semantic Analysis System (USAS)

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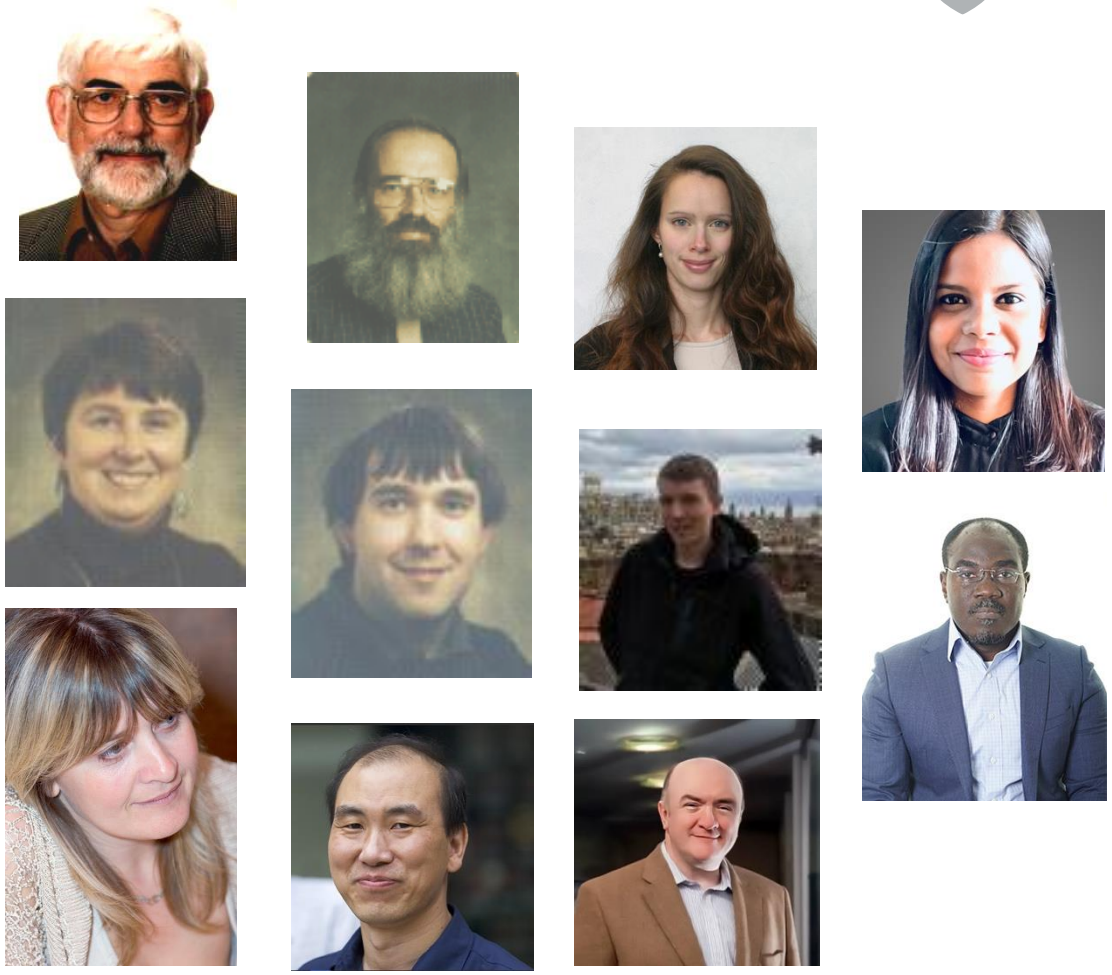
- Full text tagging, not just selected words (c.f. Diction, LIWC, RID)
- Tagging the coarse-grained sense in context, not just the word
- Not task specific categories
- Flexible category set with hierarchical structure
- Words and multi-word expressions (MWE) e.g. phrasal verbs (stubbed out), noun phrases (riding boots), proper names (United States of America), true idioms (living the life of Riley)

- https://ucrel.lancs.ac.uk/usas/
- Lexicons freely available:
  - https://github.com/UCREL/Multilingual-USAS

The work of many hands … (in Lancaster)

- Joint research with
  - Geoffrey Leech
  - Roger Garside
  - Jenny Thomas
  - Andrew Wilson
  - Dawn Archer
  - Scott Piao
  - Tony McEnery
  - Sheryl Prentice
  - Andrew Moore
  - Daisy Lal
  - Ignatius Ezeani



The work of many hands … (and beyond)

Arabic	Mo El-Haj, Elvis de Souza, Nouran Khallaf, Nizar Habash
Chinese	Richard Xiao and Qian Yufang (Lancaster), Yan Yabo, Xiaobin Yang (Hubei)
Dutch	Carole Tiberius (INL, Netherlands)
Finnish	Laura Löfberg / Johanna Vuorinen (Tampere University)
French	Michael Gauthier (Université Lumière Lyon 2, France), Emilie L'Hôte (Paris Diderot University, France), Julien Perrez, Pauline Heyvaert (Université de Liège, Belgium), Min Reuchamps (Université catholique de Louvain, Belgium), and Verena Weiland (Heidelberg, Germany)
Indonesian	Prihantoro (Lancaster)
Irish	Tim Czerniak, Gearóid Ó Donnchadha and Elaine Uí Dhonnchadha (Trinity College, Dublin), Mícheál J. Ó Meachair (Dublin City University)
Italian	Dr Francesca Bianchi (Universita del Salento, Italy) and Elena Semino (Lancaster)
Portuguese	Carmen Dayrell (Lancaster)
Russian	Olga Mudraya (Lancaster), Serge Sharoff and Bogdan Babych (Leeds)
Spanish	Ricardo-María Jiménez (Universitat Internacional de Cataluña, Barcelona, Spain) and Hugo Sanjurjo González (University of Deusto, Spain) and Carlos Herrero Zorita (Autonomous University of Madrid)
Swedish	Lisa Sjösten, Maria Nääs, Anna Gustafsson and Johan Frid (Lund University, Sweden)
Urdu	Jawad Shafi (Lancaster)
Welsh	Dawn Knight (Cardiff)

Semantic fields

- AKA concepts, semantic domains
- ‘groups together word senses that are related by virtue of their being connected at some level of generality with the same mental concept’
- Not only synonymy and antonymy but also hypernymy and hyponymy
- E.g. EDUCATION: academic, coaching, coursework, deputy head, exams, PhD, playschool, revision notes, studious, swot, viva

A General and abstract terms	**B** The body and the individual	**C** Arts and crafts	**E** Emotion
F Food and farming	**G** Government and public	**H** Architecture, housing and the home	**I** Money and commerce in industry
K Entertainment, sports and games	**L** Life and living things	**M** Movement, location, travel and transport	**N** Numbers and measurement
O Substances, materials, objects and equipment	**P** Education	**Q** Language and communication	**S** Social actions, states and processes
T Time	**W** World and environment	**X** Psychological actions, states and processes	**Y** Science and technology
Z Names and grammar			

Lexical resources for English

- Lexicon of 56,316 items
  - presentation NN1 Q2.2 A8 S1.1.1 K4
- MWE list of 18,971 items
  - travel_NN1 card*_NN* M3/Q1.2
- A small wildcard lexicon
  - *kg NNU N3.5
- Unknown words using WordNet synonym lookup

English Disambiguation methods (1)

- 1. POS tag
  - *spring* noun [season sense] [coil sense]
  - *spring* verb [jump sense]
- 2. General likelihood ranking for single-word and MWE tags
  - *green* referring to [colour] is generally more frequent than *green* meaning [inexperienced]
- 3. Overlapping MWE resolution
  - Heuristics applied: semantic MWEs override single word tagging, length and span of MWE also significant

English Disambiguation methods (2)

- 4. Domain of discourse
  - adjective *battered*
    - [Violence] (e.g. battered person)
    - [Judgement of Appearance] (e.g. battered car)
    - [Food] (e.g. battered cod)
- 5. Text-based disambiguation
  - one sense per text
- 6. Template rules
  - *Auxiliary verbs (be/do/have)*
  - *account* of NP [narrative]
  - balance of xxx *account* [financial]

Evaluation (English data)

- Hand tagged test corpus of 124,839 words
- Error rate of 8.95%
- Ambiguity ratio 47.73%
- Reduced to 17.06% by disambiguation
- Not all ambiguity is resolved, but 1st choice tag selection gives 91% accuracy.

Applications of semantic analysis 100+ papers listed at https://ucrel.lancs.ac.uk/wmatrix/

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- Analysis of market research interview transcripts
- Intelligent dictionaries
- Assistance for human translators
- Software Engineering domain understanding
- Language profiling for online child protection
- Actionability
- Corpus stylistics
- Prediction of real-world events from social media
- Metaphor and end-of-life care
- Pattern analysis of the language of psychopaths
- Political discourse analysis
- Describing the language of extremism and counter-extremism
- UK General Election Manifestos (Rayson 2008)

BRITISH NATIONAL CORPUS

ParlaMint

BNC BRITISH NATIONAL CORPUS 2014

CorCenCC Corpws Cenedlaethol Cymraeg Cyfoes National Corpus of Contemporary Welsh

Metaphor, cancer and end of life care (MELC)

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- Analysis of metaphorical language used to talk about cancer, dying and death: people 'fight' their cancer, 'win' or 'lose' their 'battle' against it, hope for a positive end to their cancer 'journey', and so on.
- 1.5M word corpus of interviews and online forum posts from patients, carers and healthcare professionals
- Methods: Manual analysis (MIP) and Wmatrix (Semantic analysis & concordancing)
- http://wp.lancs.ac.uk/melc/

G3 Warfare (e.g. *fight* as a verb, *battle*)
A1.1.1 General actions, making (e.g. *blast*, *confront*)
A1.1.2 Damaging and destroying (e.g. *destroy*, *shatter*)
E3– Violent/angry (e.g. *hit*, *attack*)
S8+ Helping (e.g. *defend*, *protect*)
S8– Hindering (e.g. *fight* as a noun)
X8+ Trying hard (e.g. *struggle*)

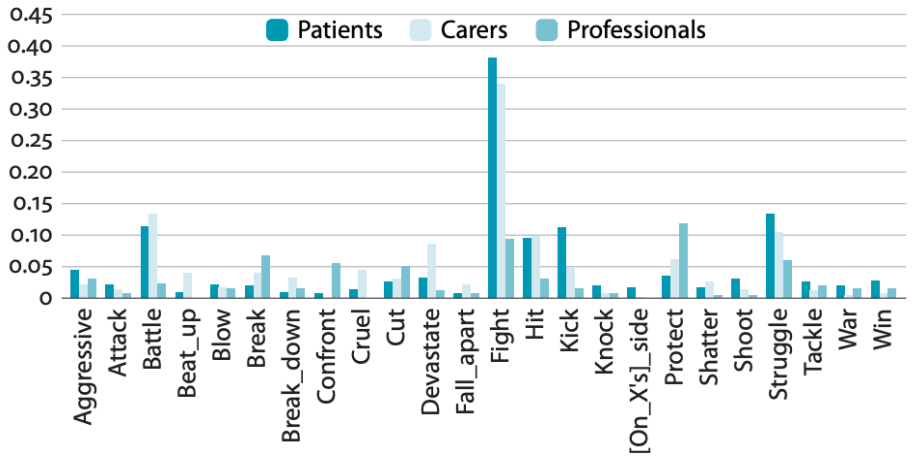

Figure 3. Relative use of most frequent Violence metaphors by each stakeholder group (per 1,000 tokens): Online forum posts

Qualitative survey analysis: FreeTxt/TestunRhydd project (2022-3)

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- Surveys are widely used in many areas of professional practice, e.g. staff development, professional training, product design, testing as well as for many types of hotel, movie and product reviews
- Very little support for bilingual free-text survey and questionnaire data analysis in English and Welsh
- Follow on funding impact project building on CorCenCC project (National Corpus of Contemporary Welsh), we will develop an open access user friendly online interface
- Partners: National Trust Wales, Cadw and National Museum Wales
- https://ucrel.lancs.ac.uk/freetxt/

Spatial Narratives project

https://spacetimenarratives.github.io

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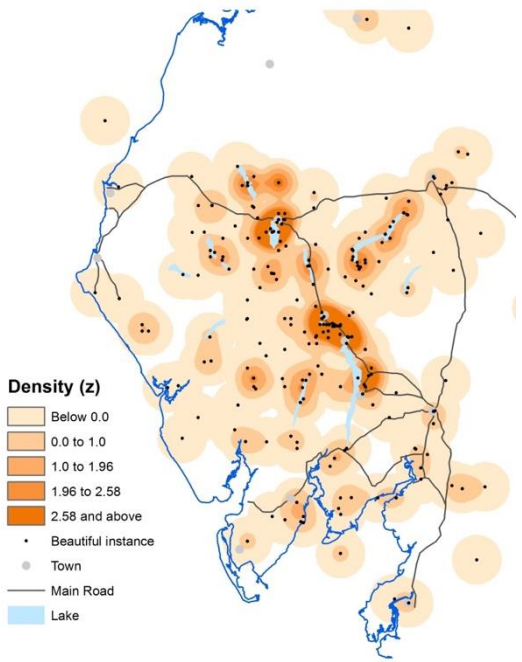
- Geographical Text Analysis (GTA): combining semantic analysis (emotions and time) with Named Entity Recognition (NER)
- What themes are associated with places? What places are associated with a theme?
- Lake District corpus & Holocaust survivor testimonies

https://github.com/SpaceTimeNarratives/demo



'Beautiful' places in 1.5 million words of Lake District writing

Named Entity Recognition applied to Lake District writing

From Penrith PLNAME two CARDINAL roads GEONOUN lead to Pooley Bridge PLNAME , about six miles QUANTITY distant, which spans the Eamont PLNAME just at its issue from Ulleswater PLNAME . Either road GEONOUN may be taken, be we recommend that which follows MOVEMENT the Shap road PLNAME to Eamont PLNAME Bridge. Carleton Hall PLNAME is near to it on the left MOVEMENT . Cross MOVEMENT the bridge GEONOUN , and take the first ORDINAL road GEONOUN to the right. At this point TIME-SEM , on the left MOVEMENT , are the druidical remains called King Arthur's Round Table PLNAME , and Mayborough PLNAME . Immediately after crossing MOVEMENT Pooley Bridge PLNAME , the road GEONOUN runs along the western shore GEONOUN of ULLESWATER PLNAME . To Patterdale PLNAME , a distance of ten miles QUANTITY ; but, before proceeding along it, the tourist MOVEMENT would do well to take a walk MOVEMENT of a few miles QUANTITY along the eastern shore GEONOUN , in the direction of Martindale PLNAME , from several points GEONOUN on which he will TIME-SEM obtain a good view of the lake GEONOUN .

Timeline

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1990-6

English semantic tagger (C version)

2002-5

Finnish semantic tagger (Java version)

2005-7

Russian semantic tagger

2013-

onwards: Extension to 10+ languages

2014-5

Historical Thesaurus Semantic Tagger (utilising HTE after 40 years of development at Glasgow University and 150+ years for the OED

2016–2021

Specialisation for genetics discourse (extension with a pre-existing Gene Ontology)

2021-2

Python code released open source (pymusas)

- https://ucrel.github.io/pymusas/
- https://pypi.org/project/pymusas/

2025

PyMUSAS integrated into Wmatrix7 (Chinese, Dutch, Finnish, French, Italian, Portuguese, Spanish, Welsh)

Recipe for creating a tagger in a new language

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1. re-evaluate USAS semantic tagset for new language context
2. find freely available (open source if possible) POS tagger & lemmatiser
3. integrate these into USAS Multilingual software framework (PyMUSAS)
   a. consider whether other new components are needed e.g. tokeniser or compound tool
4. develop single-word semantic lexicon and MWE dictionary
   a. bilingual dictionary
   b. parallel aligned corpus (Moses / Giza)
   c. machine translation / translation memory
   d. crowdsourcing by non-experts
   e. named entity recognition and gazetteers
   f. vector-based approaches
   g. multi-task & deep learning
   h. manual checking and editing by experts
5. extend disambiguation routines
6. release lexicons with CC-BY-NC-SA licence
7. release software as REST API and/or open-source licence

PyMUSAS

https://pypi.org/project/pymusas/

- Open source – Apache License Version 2.0
- Open resources – Creative Commons licence version 4
- Rule based tagger
- Identify and tag Multi Word Expressions (MWE)
- Supports multiple languages through downloadable spaCy pipelines
- Supports Indonesian and Welsh via other POS taggers (TreeTagger for Indonesian and CyTag for Welsh)

Language (BCP 47 language code)	MWE Support	Size
Mandarin Chinese (cmn)	✔	1.28MB
Welsh (cy)	✔	1.09MB
Spanish, Castilian (es)	✔	0.20MB
French (fr)	×	0.08MB
Indonesian (id)	×	0.24MB
Italian (it)	✔	0.50MB
Dutch, Flemish (nl)	×	0.15MB
Portuguese (pt)	✔	0.27MB

PyMUSAS – Language Support

Each language that we support has a guide on how to semantically tag text for that language:

https://ucrel.github.io/pymusas/

Tag Text

In this guide we are going to show you how to tag text using the PyMUSAS RuleBasedTagger so that you can extract token level USAS semantic tags from the tagged text. The guide is broken down into different languages, for each guide we are going to:

1. Download the relevant pre-configured PyMUSAS `RuleBasedTagger` spaCy component for the language.
2. Download and use a Natural Language Processing (NLP) pipeline that will tokenise, lemmatise, and Part Of Speech (POS) tag. In most cases this will be a spaCy pipeline. **Note** that the PyMUSAS `RuleBasedTagger` only requires at minimum the data to be tokenised but having the lemma and POS tag will improve the accuracy of the tagging of the text.
3. Run the PyMUSAS `RuleBasedTagger`.
4. Extract token level linguistic information from the tagged text, which will include USAS semantic tags.
5. For Chinese, Italian, Portuguese, Spanish, and Welsh taggers which support Multi Word Expression (MWE) identification and tagging we will show how to extract this information from the tagged text as well.

Chinese
Dutch
French
Italian
Portuguese
Spanish
Welsh
Indonesian

Chinese

Expand

Dutch

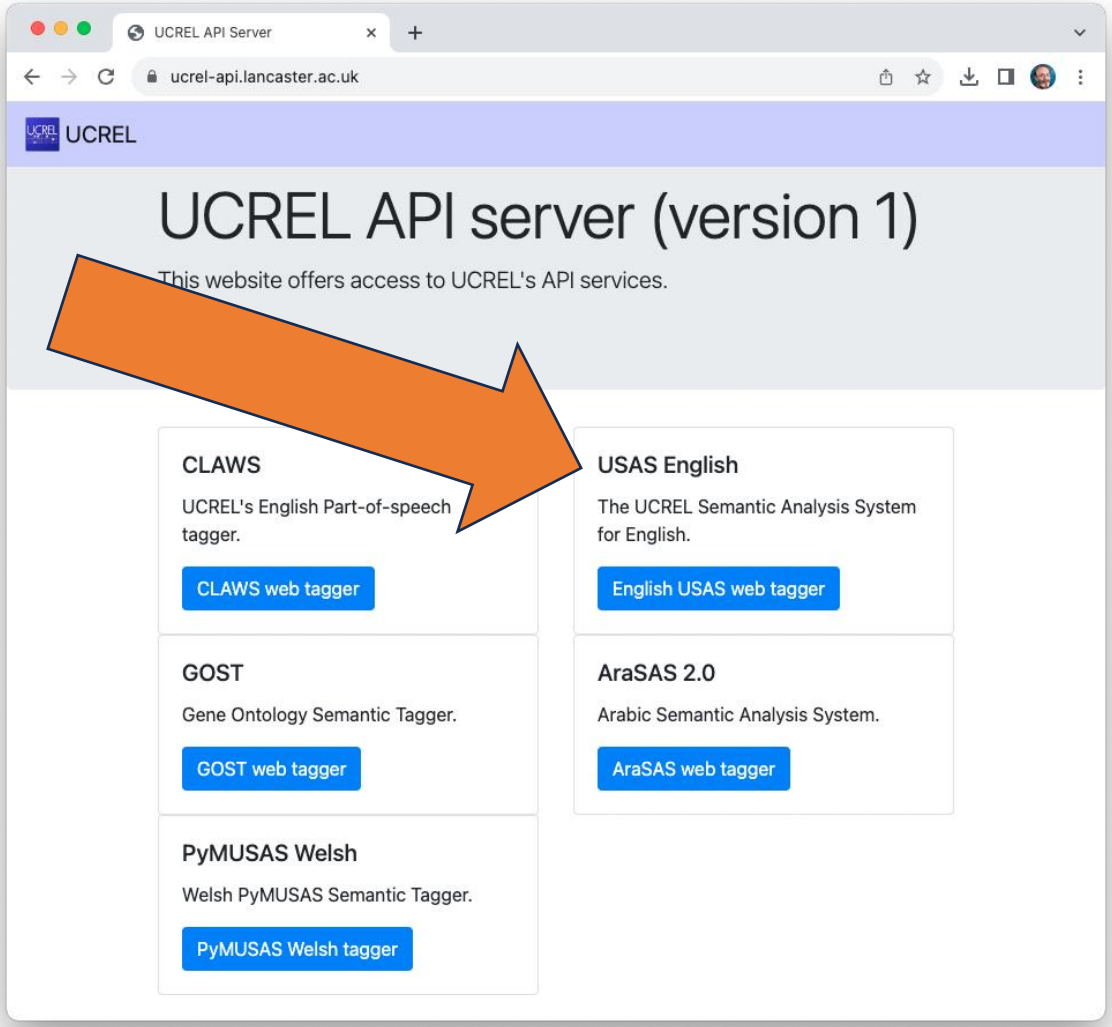
Expand

French

Expand

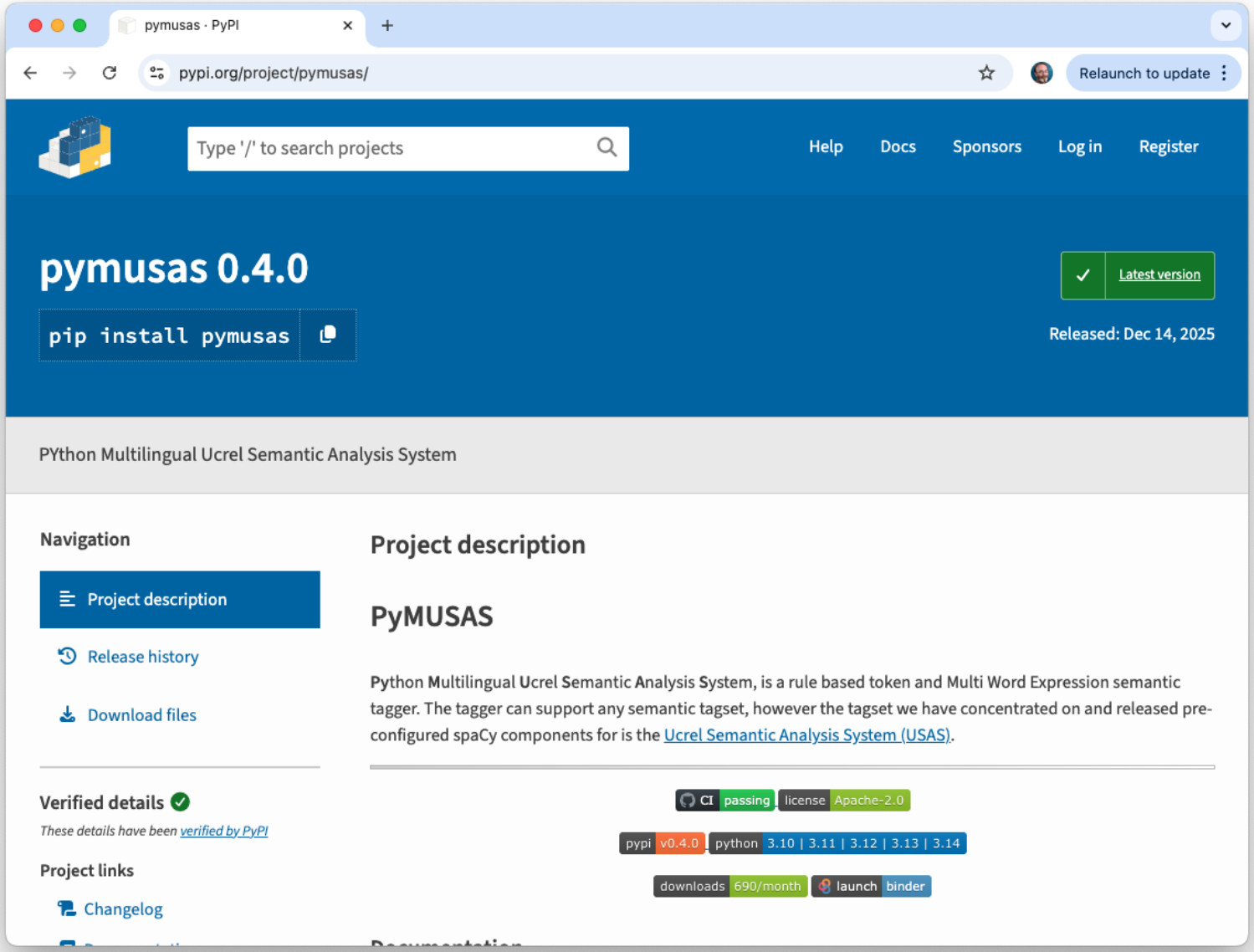
Try this Python Notebook during the lab session:
https://github.com/UCREL/pymusas_notebook

https://ucrel-api.lancaster.ac.uk/



https://pypi.org/project/pymusas/

You can also install pymusas on your own computer: the code is open source, and the lexicons are open access



Recent developments in new languages

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- Used the IgboAPI dataset (33 distinct Igbo dialects, 5,095 Igbo words with 17,979 unique dialectal word variations, complemented by 27,816 example parallel sentences) to bootstrap a lexicon for the Igbo semantic tagger
  - https://aclanthology.org/2024.lrec-main.1384/
- Creation of high quality linguistic resources (MWE lexicon) via LLMs to retrieve MWE definitions facilitating accurate translation from English to Danish lexicons, coverage evaluation and manual annotation for metaphor analysis in 4D Picture project
  - Puts et al. (2025) Pushing the boundaries: creating a Danish semantic tagger for metaphor analysis of cancer narratives. Corpus Linguistics 2025, Birmingham, UK.

Very recent developments …

- Moore et al. (2026) Creating a Hybrid Rule and Neural Network Based Semantic Tagger using Silver Standard Data: the PyMUSAS framework for Multilingual Semantic Annotation
  - largest semantic tagging evaluation of the rule based system that uses the lexical resources in the USAS framework covering five different languages (Chinese, English, Finnish, Welsh, Irish)
  - new silver labelled English corpus
  - trained and evaluated various mono and multilingual neural network models in both mono and cross-lingual evaluation setups with comparisons to their rule based counterparts, and show how a rule based system can be enhanced with a neural network model
- Updates to the Spanish lexicon (Hugo Sanjurjo, Deusto) and Danish lexicon (Sander Puts, Maastro) in January & February 2026

Deep Learning hybrid model in PyMUSAS

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- Overcomes lexical coverage limitations as it can predict a USAS tag for any word; does not require lemma or POS, just tokenization
- Bi-Encoder Model (BEM) originally created for WSD using WordNet (Belvins and Zettlemoyer 2020)
- Created an English silver labelled dataset (5.3M labelled tokens) with Wikipedia articles tagged with existing USAS tagger; train with negative sampling
- Top-1 best evaluation for Chinese (47.9), English (72.5), Finnish (60.3), Irish (57.1) and Welsh (72.4)
- Top-5 best evaluation for Chinese (70.4), English (90.1), Finnish (67.3), Irish (63.3) and Welsh (75.9)

Further reading …

- Rayson, P., Archer, D., Piao, S. L., McEnery, T. (2004). The UCREL semantic analysis system. In proceedings of the workshop on Beyond Named Entity Recognition Semantic labelling for NLP tasks in association with 4th International Conference on Language Resources and Evaluation (LREC 2004), 25th May 2004, Lisbon, Portugal, pp. 7-12.
  - http://www.lancaster.ac.uk/staff/rayson/publications/usas_lrec04ws.pdf

- Piao, S., Bianchi, F., Dayrell, C., D'Egidio, A. and Rayson, P. (2015). Development of the multilingual semantic annotation system. In proceedings of the 2015 Conference of the North American Chapter of the Association for Computational Linguistics - Human Language Technologies (NAACL HLT 2015), Denver, Colorado, United States, pp. 1268-1274.
  - http://aclweb.org/anthology/N/N15/N15-1137.pdf

Further reading …

- Piao, Scott Songlin; Rayson, Paul; Archer, Dawn; Bianchi, Francesca; Dayrell Gomes Da Costa, Maria Carmen; El-Haj, Mahmoud; Jiménez, Ricardo-María; Knight, Dawn; Křen, Michal; Lofberg, Laura; Nawab, Rao Muhammad Adeel ; Shafi, Jawad; Teh, Phoey Lee; Mudraya, Olga / Lexical Coverage Evaluation of Large-scale Multilingual Semantic Lexicons for Twelve Languages. 2016. In proceedings of the 10th edition of the Language Resources and Evaluation Conference (LREC2016), Portorož, Slovenia, pp. 2614-2619.
    - http://www.lrec-conf.org/proceedings/lrec2016/pdf/257_Paper.pdf
- El-Haj, M., Rayson, P., Piao, S., & Wattam, S. (2017). Creating and validating multilingual semantic representations for six languages: expert versus non-expert crowds. In Proceedings of the 1st Workshop on Sense, Concept and Entity Representations and their Applications: Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics. (pp. 61-71). Association for Computational Linguistics.
    - http://aclweb.org/anthology/W17-1908

Further reading …

- Andrew Moore, Paul Rayson, Dawn Archer, Tim Czerniak, Dawn Knight, Daisy Lal, Gearóid Ó Donnchadha, Mícheál Ó Meachair, Scott Piao, Elaine Uí Dhonnchadha, Johanna Vuorinen, Yan Yabo, Xiaobin Yang (2026) Creating a Hybrid Rule and Neural Network Based Semantic Tagger using Silver Standard Data: the PyMUSAS framework for Multilingual Semantic Annotation. To be presented at LREC2026.
  - https://doi.org/10.48550/arXiv.2601.09648

Summary of today's lecture

- Semantic ambiguity
- The task of Word Sense Disambiguation
  - Definitions
  - Methods
  - Evaluation
  - Resources e.g. WordNet
  - Semantic tagging via USAS
  - Applications

In other news:

Overleaf premium licence

- If you're considering using LaTeX for your SCC.453 report, then have a look at Overleaf:
- https://portal.lancaster.ac.uk/ask/digital/software/directory/overleaf/
- https://www.overleaf.com/edu/lancaster

Any final questions?

Week 18 Lab Tasks (all non-assessed)

Lancaster University

1. Automatic tagging of sample texts with USAS (approx. 30 minutes)
   - Aim: to familiarise yourself with the USAS tagset
2. Install and run the PyMUSAS semantic tagger (approx. 50 minutes)
   - Aim: to create Python code to semantically tag your texts
3. (optional) Further investigations into applications of semantic tagging for spatial humanities (approx. 30 minutes)
   - Aim: to understand how some of the semantic field annotation can be applied to other research questions

Task 1: Automatic annotation of sample texts with the USAS online tagger

Lancaster University

- Create a small corpus of sample texts e.g. news articles downloaded from today's BBC News (https://www.bbc.co.uk/news)
- Copy and paste the contents of each text one by one into the online USAS tagger (https://ucrel-api.lancaster.ac.uk/usas/tagger.html) in order to semantically tag the texts
- View the tagged output and the tag descriptions on the following pages and see if you can understand why certain words and MWEs have been tagged with these USAS labels, and can you spot places where the tags are incorrect?
  - https://ucrel.lancs.ac.uk/usas/
  - https://ucrel.lancs.ac.uk/usas/USASSemanticTagset.pdf

Task 2 Install and run the PyMUSAS semantic tagger

Lancaster University

- Either, follow the code templates in the how-to guides (https://ucrel.github.io/pymusas/) to create your own Python code to semantically tag your corpus of BBC News texts
- Or, use the existing Python Notebook and run this using Google Colab or on the SCC lab machines locally to semantically tag your texts and compare them:
  - https://github.com/UCREL/pymusas_notebook/blob/main/PyMUSAS.ipynb
- (optional) Pick a new text in a language that you don't understand e.g. Spanish and use the semantic tagger for that language to see if you can understand the main key topics in the text without using Google Translate

Task 3 Spatial Humanities applications (optional)

- In the Spatial Humanities project (https://spacetimenarratives.github.io/), we are using the semantic tagger to pick out key concepts to do with emotion (E), Movement, travel and transport (M), and time (T) to help with our research questions on extracting spatial entities from text
- There are multiple Python notebooks and a pip installable library which you can test out to see how this works:
  - https://github.com/SpaceTimeNarratives/demo
  - https://github.com/SpaceTimeNarratives/spatio-textual